ECON 5040 Midterm: Comprehensive Study Guide

Dynamic Stochastic Modeling (Fisher), Fall 2026. Covers all 10 topics on the official study guide, in Fisher's notation. The exam is closed book: one question per topic, answered in 3–4 sentences plus maybe a graph. The worked answers are in `ECON5040_Sample_Midterm.pdf`.

How to write a full-credit answer (every topic)

(1) State the central idea in one sentence. **(2)** Write the key equation and say what *each side* means economically. **(3)** Give the mechanism or comparative static ("if X rises, then … because …"). **(4)** Add a labeled graph if one exists for the topic. Fisher says *not* to memorize long derivations or code; he grades whether you can interpret the equations and explain how the pieces fit together.

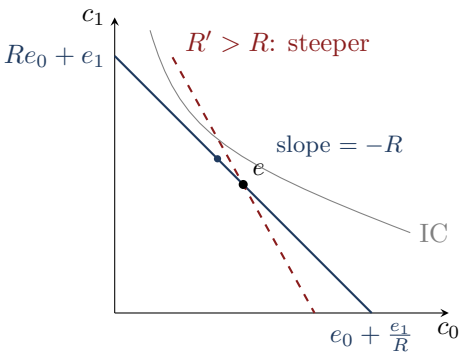
1. Inter-temporal Prices

Big idea

Consumption today (c_0) and consumption next year (c_1) are *different commodities*. The gross real interest rate $R = 1 + r$ is the relative price between them: give up 1 unit today and you get R units tomorrow.

Prices. Set today's good as the numéraire, $p_0 = 1$. The price today of one unit of future consumption is $p_1 = 1/R$, so $R = p_0/p_1$: the price of current consumption in units of future consumption.

Budget. With endowment (e_0, e_1): $c_0 + \dfrac{c_1}{R} = e_0 + \dfrac{e_1}{R}$ (present value of wealth). The budget line has slope $-R$ and passes through the endowment.



Optimality (interior). MRS = relative price:

$$\frac{u'(c_0)}{\beta u'(c_1)} = R.$$

The left side is how many units of c_1 you need to give up one unit of c_0; the right side is how many the market gives you.

Substitution incentive. A rise in R makes future consumption *cheaper* (its price $1/R$ falls) and current consumption more expensive. The substitution effect therefore always pushes the household to consume less today and save more. Graphically, the budget line pivots *through the endowment* and gets steeper. With CRRA utility, $c_1/c_0 = (\beta R)^{1/\rho}$, so $1/\rho$ measures how strongly the household substitutes (the elasticity of intertemporal substitution).

Income/wealth effect (for completeness). A lender (with $c_0 < e_0$) becomes richer when R rises, which pushes c_0 up. A borrower becomes poorer. So the *total* effect on saving is ambiguous for a lender, but the substitution effect alone is unambiguous.

Time separability. $U = \sum_t \beta^t u(c_t)$. Here β is impatience, and u ("felicity") has the same shape every period. Separability means the MRS between t and $t+1$ does not depend on consumption at other dates (Fisher's breakfast/lunch/dinner example). It is restrictive but tractable.

Traps / what graders look for

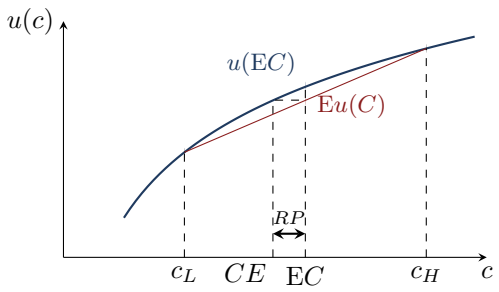
Say "relative price" explicitly. R is a price, not just a return. A higher R makes the *future* cheap. Don't claim saving *must* rise; that holds for the substitution effect only.

2. Choice Under Uncertainty

Big idea

Under the vNM axioms (completeness, transitivity, continuity, independence), lotteries are ranked by **expected utility** $\mathrm{E}[u(C)] = \sum_s \pi_s u(c_s)$. The *curvature* of u encodes the attitude toward risk.

Risk aversion. u concave ($u'' < 0$) $\iff$ $u(\mathrm{E}C) \geq \mathrm{E}u(C)$ (Jensen). Marginal utility is diminishing, so a bad outcome hurts more than an equal-sized good outcome helps.
Certainty equivalent. $u(CE) = \mathrm{E}u(C)$, so $CE = u^{-1}(\mathrm{E}u(C))$. This is the sure amount the consumer values the same as the gamble. It satisfies $CE < \mathrm{E}C$ when the consumer is risk averse.
Risk premium. $RP = \mathrm{E}C - CE$: the expected consumption the consumer would give up to remove the risk.



Canonical example (memorize). $u = \log c$, with $C = 50$ or 150 each with probability $\frac{1}{2}$. Then $\mathrm{E}C = 100$, and $\log CE = \frac{1}{2}\log 50 + \frac{1}{2}\log 150 \Rightarrow CE = \sqrt{50 \cdot 150} = \sqrt{7500} \approx \mathbf{86.60}$, so $RP \approx 13.40$.
General recipe: (1) compute $\mathrm{E}u$; (2) invert u. With log utility, CE is the *geometric mean* of the outcomes.

Curvature measures. The coefficient of absolute risk aversion is $A(c) = -u''/u'$ (for fixed-dollar gambles). The coefficient of relative risk aversion is $\rho(c) = -cu''/u'$ (for percentage gambles). Under CRRA, $u = \frac{c^{1-\rho}-1}{1-\rho}$, ρ is constant; log is the case $\rho = 1$, with $A = 1/c$ (decreasing). More curvature gives a lower CE and a bigger RP. A risk-neutral consumer (linear u) has $CE = \mathrm{E}C$.

vNM vs. ordinary utility. vNM utility is unique only up to a *positive affine* transformation $a + bu$; an arbitrary increasing transformation would change how lotteries are ranked.

Traps / what graders look for

Don't compute $u(\mathrm{E}C)$ and call it the CE. The CE comes from inverting *expected* utility. Always state the inequality $CE < \mathrm{E}C$ and say it is *because* u is concave.

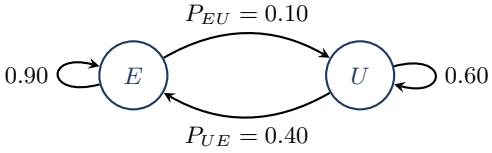
3. Markov Processes

Big idea

Markov property: $\Pr(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_0) = \Pr(X_{t+1} = j \mid X_t = i)$. Once you know today's state, the earlier history adds nothing to the forecast of tomorrow. History still matters, but *only through* the current state.

Transition matrix. $P_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$. Rows are today's state and columns are tomorrow's.

- **Element P_{ij}:** the probability of moving to j next period given you are in i now. Example: with $P = \begin{bmatrix} .9 & .1 \\ .4 & .6 \end{bmatrix}$ (E, U), $P_{UE} = 0.4$ is the *job-finding rate* and $P_{EU} = 0.1$ is the *job-separation rate*.
- **Row i:** the entire conditional distribution of tomorrow's state given i today. It is non-negative and sums to 1 (a stochastic matrix).



Distributions. Write the distribution as a row vector ψ_t. It evolves by $\psi_{t+1} = \psi_t P$, so $\psi_t = \psi_0 P^t$, and $(P^n)_{ij} = \Pr(X_{t+n} = j \mid X_t = i)$. For example, $(P^2)_{EU} = .9(.1) + .1(.6) = 0.15$.
Stationary distribution. $\psi^* = \psi^* P$. In the example, $u = .1(1-u) + .6u \Rightarrow u^* = 0.2$, so $\psi^* = (0.8, 0.2)$. Individuals keep moving between states, but the *shares* stay constant. Long-run simulated frequencies approach ψ^* (LLN).

Why it matters later. In a DP, the state's transition is a Markov chain that can depend on the action (M_u). Once you fix an optimal policy h, it induces its own chain $P_h(x, \cdot) = M_{h(x)}(x, \cdot)$.

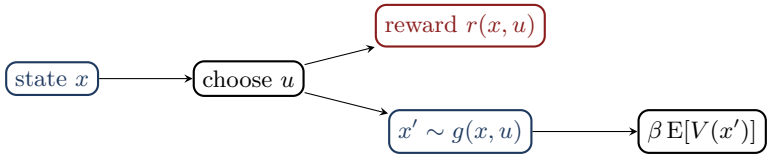
4. Introduction to Dynamic Programming

Big idea

"What we choose now changes our future irrevocably." The **Bellman equation** splits an infinite-horizon problem into *today's reward* plus the *discounted expected value of where you end up*:

$$V(x) = \max_{u \in U} \Big\{ \underbrace{r(x,u)}_{\text{reward today}} + \beta \underbrace{\mathrm{E}[V(x') \mid x, u]}_{\text{continuation value}} \Big\}, \qquad x' \sim g(x,u).$$

The five ingredients: the state space X, the control set U, the reward $r(x,u)$, the (random) transition $x' \sim g(x,u)$, and the discount factor $\beta \in [0,1)$. The original problem is $\max \mathrm{E} \sum_t \beta^t r(x_t, u_t)$. The Bellman equation replaces choosing an infinite sequence with choosing *one* action today, given V. The catch is that V appears on both sides, so it is a *fixed-point* equation.



How to identify the pieces (practice table).

Problem	State x	Choice u	Current payoff	Next state x'
Optimal growth	k_t	k_{t+1}	$\log(Ak^\alpha - k')$	$k' = u$
Cake eating	x_t (cake)	c_t	$u(c)$	$x' = x - c$
Student TV/Study	health	TV / Study	$r(x,u)$ table	$x' \sim M_u(x,\cdot)$
McCall search	offer w	accept / reject	w or c	$W' \sim F$ (if reject)
Income fluctuation	(a,z)	a' (or c)	$u(Ra + y(z) - a')$	(a', z'), $z' \sim P(z,\cdot)$

The state is *everything you need today* to make the decision and to forecast tomorrow. It is a "sufficient statistic" for the past, which is the Markov idea. **The choice** is what the agent controls. The control can affect today's reward, tomorrow's state, or both.

Worked example (optimal growth). $V(k) = \max_{k'}\{\log(Ak^\alpha - k') + \beta V(k')\}$. Guess $V = E + F\log k$ and verify: $F = \alpha/(1-\alpha\beta)$, giving the policy $k' = \alpha\beta A k^\alpha$ (save the constant fraction $\alpha\beta$ of output). The steady state is $k^* = (\alpha\beta A)^{1/(1-\alpha)}$.

5. Value Functions and Policy Functions

Big idea

Value function $V(x)$: "How good is it to be here right now, knowing my future plans are as good as can be?" It is a *number* for each state (lifetime utility).
Policy function $h(x)$: "What should I do, come what may?" It is an *action* for each state (a contingent plan).

- In a stochastic world the solution cannot be a fixed sequence $(u_0, u_1, \dots)$, because future states are unknown. It has to be a *rule*, $u_t = h(x_t)$.
- **The link:** the policy is the argmax of the Bellman equation,

  $$h(x) = \arg\max_u \{r(x,u) + \beta \mathrm{E}[V(x') \mid x, u]\},$$

  and the value is what you get by following it: $V(x) = r(x, h(x)) + \beta \mathrm{E}[V(x') \mid x, h(x)]$. Know V and you get h from one maximization. Know h and you get V_h by solving a linear system.
- **Economic information:** V measures welfare and the value of resources. $V'(x)$ is the shadow value of one more unit of the state; by the envelope condition $V'(x) = u'(c)$ in cake eating and $u'(c)f'(k)$ in growth. h gives *behavior*: the saving rule, the reservation-wage rule, the law of motion $k' = h(k)$, and the Markov chain P_h that drives long-run outcomes.

- Examples: in cake eating $V(x)$ is maximal lifetime utility and $c^*(x) = (1-\beta^{1/\rho})x$; in McCall $V(w) = \max\{A(w), C\}$ and $h(w) =$ accept iff $w \geq \bar{w}$; in growth $h(k) = \alpha\beta Ak^\alpha$.

Chain to remember: dynamic problem → Bellman equation → value function → policy function → law of motion / Markov chain → long-run outcomes.

6. Howard's Method versus Value Function Iteration

Big idea

Both methods solve the same Bellman equation. **VFI** starts from a guessed *value function* and iterates $V_{n+1} = TV_n$. **Howard** starts from a guessed *policy* and alternates *exact policy evaluation* with *policy improvement*.

VFI. The Bellman operator is $(TV)(x) = \max_u\{r(x,u) + \beta\sum_{x'} M_u(x,x')V(x')\}$. Start at $V_0 = 0$, so $V_1 = \max_u r(x,u)$ (only today counts). Each application of T adds one more period of future rewards. Stop when $\|V_{n+1} - V_n\|$ is tiny, then read off $h = \arg\max$. T is a contraction with modulus β, so the error shrinks by a factor $\approx \beta$ per step. That is *slow when β is close to 1*: hundreds of iterations, each requiring a full maximization.

Howard (policy iteration).

1. Guess a policy h_0.
2. **Evaluate:** with the policy fixed, the max disappears and $V_h = r_h + \beta P_h V_h \Rightarrow V_h = (I - \beta P_h)^{-1} r_h$. This is one linear solve.
3. **Improve:** $h_1(x) = \arg\max_u Q(x,u;V_{h_0})$, where $Q = r(x,u) + \beta\sum M_u(x,x')V_{h_0}(x')$.
4. Stop when the policy doesn't change; otherwise repeat.

Why Howard is faster. The evaluation step computes the value of following h *forever* in one shot, which is equivalent to infinitely many VFI steps with the policy held fixed. Each improvement step then makes a big jump. There are only finitely many policies ($|U|^{|X|}$, e.g. $2^3 = 8$ for the student) and each round weakly improves, so it converges in a *handful* of iterations. The trade-off is that each iteration costs a linear solve.

Student example ($\beta = .5$, $h_0 =$ TV everywhere): $V_{h_0} \approx (3.64, 6.92, 13.40)$. The improvement check gives TV in every state (e.g. Sick: TV 3.64 vs. Study 0.98), so the guess was already optimal and the method stops after one round.

Traps / what graders look for

Key words: VFI = "value guess, apply T until V converges." Howard = "policy guess, *evaluate* (a linear system), *improve* (argmax), stop when the policy is unchanged."

7. The Solow Model

Big idea

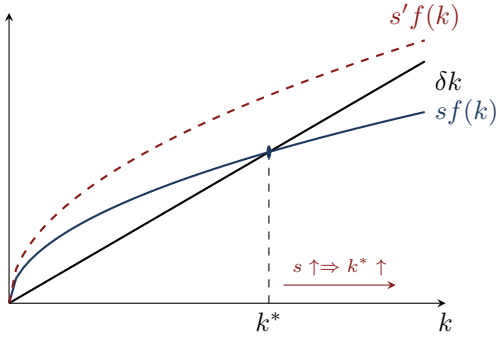
The saving rate s is **exogenous**. Capital accumulates according to $k_{t+1} = sf(k_t) + (1-\delta)k_t$. Diminishing returns ($f'' < 0$) mean investment eventually only covers depreciation, so the economy converges to a **steady state** $sf(k^*) = \delta k^*$.

Steady state. Investment equals replacement: $sf(k^*) = \delta k^*$. With $f = Ak^\alpha$,

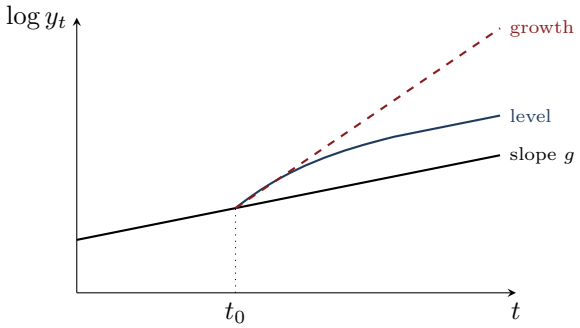
$$k^* = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}, \quad y^* = A(k^*)^\alpha, \quad c^* = (1-s)y^*.$$

Convergence. If $k < k^*$ then $sf(k) > \delta k$ and k rises; if $k > k^*$ it falls. In the 45° diagram, k^* is where $g(k)$ crosses the 45° line.

Comparative statics. $s \uparrow$, $A \uparrow \Rightarrow k^*, y^* \uparrow$; $\delta \uparrow$ (or $n \uparrow$) $\Rightarrow k^* \downarrow$.



Level vs. growth (the key distinction). A permanent rise in s raises k^* and y^*: a **level effect**. Growth is temporarily faster during the transition, but at the new steady state, per-worker growth returns to its old rate. Long-run growth of output per worker equals the rate of technical progress g, which comes from $E_{t+1} = (1+g)E_t$ (aggregate output grows at $n + g + ng$). Saving cannot permanently raise growth because diminishing returns choke off the extra investment. Only a change in g (or ongoing increases in A) changes the *slope* of $\log y_t$.



Extras.

- **Golden rule.** More saving need not raise $c^* = (1-s)f(k^*)$. With Cobb–Douglas, $s_{GR} = \alpha$, but Solow doesn't explain why households would pick it.
- **Factor prices.** $r = \alpha A k^{\alpha-1} - \delta$ falls with k, and $w = (1-\alpha)Ak^{\alpha}$ rises with k.
- **Convergence speed.** $\lambda = 1 - \frac{(1-\alpha)(\delta+n+g+ng)}{(1+n)(1+g)}$, half-life $= \ln(1/2)/\ln\lambda \approx 13$ years. Note that s does not affect the speed.

8. Optimal Saving

Big idea

Saving is now a *choice*. The **Euler equation**

$$\underbrace{u'(c_t)}_{\text{MU of eating 1 more unit today}} = \underbrace{\beta R\, \mathrm{E}_t[u'(c_{t+1})]}_{\text{discounted expected MU of saving it: } R \text{ units tomorrow}}$$

says that at the optimum, you can't gain by shifting one unit between today and tomorrow.

Derivation logic (cake eating, $R = 1$). $V(x) = \max_c\{u(c) + \beta V(x-c)\}$. The FOC is $u'(c_t) = \beta V'(x_{t+1})$ and the envelope condition is $V'(x) = u'(c)$. Together these give $u'(c_t) = \beta u'(c_{t+1})$. With CRRA, $c_{t+1}/c_t = (\beta R)^{1/\rho}$, and in cake eating the policy is $c^*(x) = (1-\beta^{1/\rho})x$ ($= (1-\beta)x$ for log). In the growth model, R is replaced by $f'(k_{t+1})$: MRS = marginal efficiency of investment = gross real interest rate.

When the two sides are not equal (the exam question).

- $u'(c_t) > \beta R \mathrm{E}u'(c_{t+1})$: a unit eaten today is worth more than the same unit saved. The household should **consume more today and save less**. That lowers $u'(c_t)$ and raises $u'(c_{t+1})$ (since c_{t+1} falls) until the two sides are equal.
- $u'(c_t) < \beta R \mathrm{E}u'(c_{t+1})$: saving is worth more, so the household should **consume less today and save more**, until equality.
- Exception: with a borrowing constraint $a' \geq \underline{a}$ that binds, the household *wants* to consume more but can't, so $u'(c_t) > \beta R \mathrm{E}u'(c_{t+1})$ persists at the optimum (the Euler *inequality*).

Comparative statics. $\beta \uparrow$ (more patient) means save more. $R \uparrow$ means a steeper consumption profile (substitution). If $\beta R < 1$, consumption declines over time; if $\beta R = 1$, it is flat (perfect smoothing).

9. Income Fluctuations

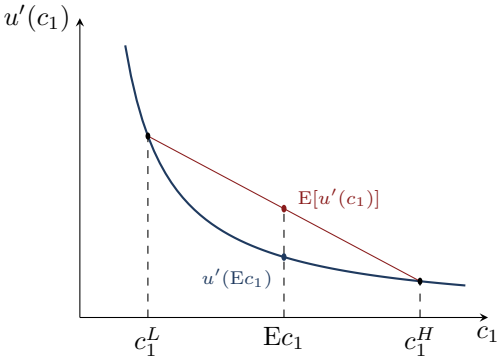
Big idea

With uninsurable income risk and incomplete markets, a risk-free asset is the only tool. Households **self-insure** by building a buffer of assets. **Precautionary saving** means uncertainty itself raises saving, even when *expected* income is unchanged.

Why (math). Euler: $u'(c_0) = \beta R \mathrm{E}[u'(c_1)]$. If u' is *convex* ($u''' > 0$, e.g. log, where $u' = 1/c$), Jensen gives $\mathrm{E}[u'(c_1)] > u'(\mathrm{E}c_1)$. A mean-preserving spread in Y_1 therefore raises the right side at any given a_1. To restore equality the household must raise $u'(c_0)$, which means it must **save more**.

Why (economics). Bad states have very high marginal utility, so an extra dollar of assets is especially valuable when income turns out low.

Classroom example: $Y_1 \in \{2, 10\}$ vs. a certain 6, with the same mean; the risky case saves more.



Self-insurance. The asset pays R in *every* state, so it is not state-contingent insurance. But consumption is chosen *after* income is observed, so assets can be drawn down when income is low. The buffer transfers resources toward the future states where consumption and wealth are low. Three distinct ideas: *smoothing* (dislike fluctuations), *self-insurance* (the buffer), and *precautionary saving* (risk raises saving relative to the no-risk benchmark).

Infinite-horizon problem. $V(a, z) = \max_{a' \geq \underline{a}} \{u(Ra + y(z) - a') + \beta \sum_{z'} P(z, z') V(a', z')\}$. The state is (a, z): assets summarize the past, and z forecasts future income when income is persistent (Markov). The control is a'. The borrowing constraint turns the Euler equation into $u'(c_t) \geq \beta R \mathrm{E}_t u'(c_{t+1})$, with $>$ in states where the constraint binds. This model combines **DP + Markov chains + optimal saving**.

10. Putting the Ideas Together: Job Search (McCall)

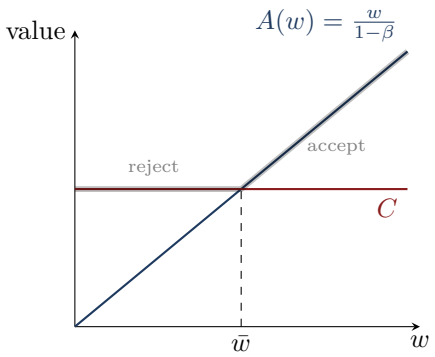
Big idea

Job search is **dynamic**: accepting gives w forever, while rejecting gives c today *plus the option* to draw again tomorrow. Today's choice affects both today's payoff and tomorrow's opportunities. It is an **optimal stopping** problem: the question is *when to stop*, not which wage to choose.

Formulation.

- State: the current offer $w \in [w_{\min}, w_{\max}]$, a continuum, since Nature draws w.
- Choice: {Accept, Reject}.
- Reward: w forever if employed; c (unemployment compensation) if searching.
- Transition: if you reject, $W' \sim F$ IID; if you accept, you are absorbed (the job is permanent).
- Discount: β.

$$V(w) = \max \left\{ \underbrace{\tfrac{w}{1-\beta}}_{A(w)}, \quad \underbrace{c + \beta \mathrm{E}[V(W')]}_{C \text{ (continuation value)}} \right\}.$$



Reservation wage. $A(w)$ is increasing in w. C is *constant* in w, because offers are IID and today's offer doesn't change tomorrow's distribution. They cross once at $\bar{w}$, where $\frac{\bar{w}}{1-\beta} = C \Rightarrow \bar{w} = (1-\beta)[c + \beta \mathrm{E}V(W')]$. The policy is to accept iff $w \geq \bar{w}$, so a single number summarizes the whole policy.

Comparative statics. $c \uparrow \Rightarrow C \uparrow \Rightarrow \bar{w} \uparrow$ (longer search). $\beta \uparrow$ (more patient) $\Rightarrow$ waiting is worth more $\Rightarrow \bar{w} \uparrow$. A mean-preserving spread in F raises $\bar{w}$, because you keep the upside and reject the downside (option value).

How it ties the course together: the Bellman equation (Topic 4), value vs. policy (Topic 5, $V(w)$ vs. the $\bar{w}$ rule), VFI to compute C (Topic 6), and expected values over a distribution (Topic 2). With separations, employment status becomes part of the state and the E/U dynamics form a Markov chain (Topic 3).

One-line recall for each topic

1 R = price of c_0 in units of c_1; $R \uparrow$ makes the future cheap, so substitute toward saving.
2 Eu; concave $\Rightarrow CE <$ EC; $CE = u^{-1}(\mathrm{E}u)$; log 50/150 gives 86.60.
3 The present is a sufficient statistic; $P_{ij} = \Pr(j \text{ tomorrow} \mid i \text{ today})$; a row is a full conditional distribution.
4 $V = \max_u\{r + \beta \mathrm{E}V(x')\}$; state / choice / payoff / transition / β.
5 V = how good it is to be here; h = what to do here; h = arg max of the Bellman equation.
6 VFI: $V_{n+1} = TV_n$. Howard: evaluate $(I - \beta P_h)^{-1} r_h$, improve, few iterations.
7 $sf(k^*) = \delta k^*$; $s \uparrow$ is a level effect; long-run growth = g (technology).
8 $u'(c_t) = \beta R \mathrm{E}u'(c_{t+1})$; LHS > RHS $\Rightarrow$ consume more now, LHS < RHS $\Rightarrow$ save more.
9 Convex $u' \Rightarrow$ risk raises E$u'(c_1) \Rightarrow$ save more; assets are a buffer for bad states.
10 State w, accept/reject, $A(w) = w/(1-\beta)$ vs. constant C, reservation wage $\bar{w}$.